

PORTLAND HP, United Kingdom

WMO: 038580

Lat: 50.565N

Lon: 2.449W

Elev: 3

StdP: 101.29

Time Zone: 0.00 (EUW)

Period: 88-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.6	1.1	-4.7	2.5	2.0	-3.1	2.9	3.1	18.3	9.0	16.1	8.8	8.4	50	0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.7	22.9	17.4	21.3	16.7	20.1	16.3	18.3	21.0	17.7	20.0	17.2	19.3	4.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.4	12.4	19.1	16.8	12.0	18.6	16.2	11.5	18.2	51.6	21.3	49.7	20.0	48.2	19.4	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.4	13.8	12.4	-2.2	25.5	2.5	2.4	-4.0	27.2	-5.4	28.6	-6.8	30.0	-8.6	31.7		
(5)			-3.1	20.2	2.5	1.4	-4.9	21.2	-6.3	22.0	-7.7	22.8	-9.5	23.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	7.2	6.9	8.3	9.2	12.6	14.7	16.7	17.5	15.4	12.9	9.8	7.7		
	DBStd	4.37	2.81	2.72	1.79	1.95	2.57	1.94	1.67	1.81	1.60	2.47	2.76	3.06		
	HDD10.0	398	88	87	55	37	5	0	0	0	0	7	37	82		
	HDD18.3	2472	344	319	309	275	179	111	56	37	87	168	256	331		
	CDD10.0	988	2	1	3	12	84	141	209	233	163	97	32	9		
	CDD18.3	22	0	0	0	0	1	2	6	12	1	0	0	0		
	CDH23.3	35	0	0	0	0	1	4	7	23	1	0	0	0		
	CDH26.7	2	0	0	0	0	0	0	0	2	0	0	0	0		
(14) Wind	WSAvg	6.2	7.7	6.9	6.4	5.6	5.1	5.0	5.0	5.0	5.9	7.1	6.5	7.8		
(15) Precipitation	PrecAvg	836	96	71	61	54	52	50	45	58	63	93	100	94		
	PrecMax	1127	194	165	130	137	111	122	122	129	129	141	207	173		
	PrecMin	667	19	11	3	2	8	8	6	9	13	33	24	29		
	PrecStd	119	42	39	26	33	28	28	27	32	32	32	44	44		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	12.2	13.8	16.5	22.2	23.5	24.4	25.7	21.2	18.7	15.1	13.2		
		MCWB	11.1	10.7	10.8	11.8	15.7	16.7	18.2	18.2	17.1	15.8	13.6	11.9		
	2%	DB	11.4	11.2	12.0	14.2	19.9	21.0	22.3	23.3	19.5	17.3	14.5	12.6		
		MCWB	10.5	10.1	9.7	10.8	14.8	15.5	17.2	17.9	16.0	14.9	13.2	11.5		
	5%	DB	11.0	10.7	11.2	12.8	18.0	19.1	20.9	21.3	18.7	16.6	14.0	12.2		
		MCWB	10.2	9.5	9.3	9.9	13.5	14.7	16.7	17.3	15.6	14.5	12.9	11.2		
	10%	DB	10.6	10.2	10.6	11.8	16.3	17.7	19.7	20.1	17.9	16.0	13.5	11.7		
		MCWB	9.7	9.0	9.0	9.5	12.9	14.2	16.1	16.8	15.2	14.2	12.3	10.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	11.2	11.2	12.7	16.8	17.3	19.1	19.8	18.1	16.3	14.2	12.6		
		MCDB	11.7	11.6	12.5	15.3	20.7	22.4	22.7	22.8	19.9	17.4	14.7	12.9		
	2%	WB	10.9	10.6	10.5	11.3	15.2	16.1	18.0	18.6	17.2	15.7	13.7	12.0		
		MCDB	11.2	11.0	11.5	13.1	18.9	19.6	20.9	21.4	18.4	16.8	14.2	12.4		
	5%	WB	10.5	9.9	10.0	10.6	14.1	15.3	17.2	18.0	16.7	15.2	13.1	11.5		
		MCDB	10.9	10.5	10.7	12.0	17.3	18.2	19.7	20.1	17.9	16.2	13.9	12.0		
	10%	WB	9.9	9.2	9.6	10.1	13.2	14.6	16.6	17.5	16.1	14.6	12.6	10.8		
		MCDB	10.5	10.0	10.3	11.3	15.8	17.0	18.8	19.6	17.4	15.8	13.4	11.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.2	3.5	3.6	4.6	5.1	4.7	4.4	4.7	4.4	3.8	3.5	3.3		
		MCDBR	2.9	3.6	4.0	6.0	7.6	7.4	6.7	6.5	5.0	3.9	3.0	2.8		
	5% WB	MCWBR	3.3	3.5	3.2	3.9	4.2	4.2	3.9	3.7	3.5	3.4	3.1	3.2		
		MCWBR	3.0	3.4	3.1	4.8	6.8	6.4	5.5	5.1	3.9	3.6	3.0	2.9		
(40) Clear-Sky Solar Irradiance	taub		0.335	0.350	0.385	0.400	0.403	0.396	0.390	0.383	0.374	0.363	0.335	0.325		
	taud		2.375	2.386	2.292	2.280	2.317	2.369	2.389	2.430	2.440	2.438	2.425	2.352		
	Ebn at Noon		685	778	809	841	855	862	861	850	818	753	682	632		
	Edh at Noon		65	82	107	121	122	117	112	103	92	78	62	58		
(44) All-Sky Solar Radiation	RadAvg		0.82	1.56	2.79	4.39	5.31	5.76	5.47	4.58	3.45	1.93	1.03	0.65		
	RadStd		0.07	0.17	0.35	0.39	0.28	0.57	0.40	0.36	0.22	0.14	0.10	0.08		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page